

Lipofectamine 3000 protocol

PLASMID / CONDITION	NG/ML	WELLS
EV	823	4.5
WT	718	4.5
MUT	479	4.5
13.5 wells total		

- Seed cells in a **12 well plate**
 - Cell type: Caco-2
- No. of cells: 1×10^5 (typical $0.5\text{--}2 \times 10^5$)
- Cells should be **70–90 %** confluent.
- Work in the dark
- Reduce the medium volume in each well to **0.5 mL** to improve transfection efficiency.
- Prepare two mixes
 - Optimem and Lipofectamine (official range 1.5–3 μL per μg DNA, this sheet uses the low end)
 - 50 μL Optimem per well 675 μL
 - 1.5 $\mu\text{L}/\mu\text{g}$ DNA $\rightarrow$ **0.75 μL** Lipofectamine per well 10.13 μL
 - Optimem, DNA and P3000 Reagent – add P3000 last (change the DNA and Lipofectamine and P3000 follow)
 - 50 μL Optimem per well 675 μL
 - 500 ng DNA per well 6750 ng
 - 2 $\mu\text{L}/\mu\text{g}$ DNA $\rightarrow$ **1 μL** P3000 per well 13.5 μL
- Mix well (no need to vortex)
- Add DNA Mix to the Lipofectamine mix
- Incubate 10–15 min and drop-wise **100 μL** mix per well.
- Check cells in microscope
- Spin down plate
- ~~Change medium at the end of the day (2.5 h minimum)~~
- Incubate cells for 2–4 days at 37 °C

EV 4.5 wells

 225 μL Optimem
3.375 μL Lipofectamine

 225 μL Optimem
2.73 μL DNA
4.5 μL P3000

WT 4.5 wells

 225 μL Optimem
3.375 μL Lipofectamine

 225 μL Optimem
3.13 μL DNA
4.5 μL P3000

MUT 4.5 wells

 225 μL Optimem
3.375 μL Lipofectamine

 225 μL Optimem
4.7 μL DNA
4.5 μL P3000

$$EV: 500 \text{ ng} \times 4.5 \div 823 \text{ ng}/\mu\text{L} = 2.73 \mu\text{L}$$

$$MUT: 500 \text{ ng} \times 4.5 \div 479 \text{ ng}/\mu\text{L} = 4.7 \mu\text{L}$$

$$WT: 500 \text{ ng} \times 4.5 \div 718 \text{ ng}/\mu\text{L} = 3.13 \mu\text{L}$$

$$\text{cells: } 1 \times 10^5 \times 13.5 \text{ wells} = 1.35 \times 10^6 \text{ [seeding tool](#)}$$